APPLICATION SOURCE CODE DOCUMENT

Smart Study Planner — Implementation Source

Self-contained responsive web application for task tracking, analytics, and priority-based scheduling.

ARCHITECTURE	PERSISTENCE	UI GRID	SECURITY
Single-Page Vanilla JS	Browser LocalStorage API	CSS Grid & Flexbox	HTML Entity Escaped

Module Component	Responsibilities	Key Implementation Details
Analytics Engine	Real-time study metric computation	Aggregates task counters, completion ratios, and dynamic progress fill bar.
Input Controller	Schedule creation & validation	Captures subject, topic, date, start time, duration, and priority level.
Heuristic Scheduler	Automated schedule generation	Sorts by priority matrix with automated 10-minute breaks between study blocks.

Part 1: HTML Head & Complete CSS Stylesheet

<style> block

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>Smart Study Planner</title> <style> *{box-sizing:border-box} body{margin:0;font-family:Arial,sans-serif;background:#f3f5fb;color:#202641} header{background:#5146e5;color:white;padding:22px 7%;display:flex;justify-content:space-between;align-items:center;flex-wrap:wrap;gap:12px} header h1{margin:0;font-size:25px} header p{margin:7px 0 0;font-size:14px} main{max-width:1150px;margin:25px auto;padding:0 18px} .stats{display:grid;grid-template-columns:repeat(auto-fit,minmax(180px,1fr));gap:15px;margin-bottom:22px} .card,.panel{background:white;padding:20px;border-radius:14px;box-shadow:0 3px 15px #2026410d} .stat-label{color:#777;font-size:14px} .stat-number{font-size:29px;font-weight:bold;margin-top:10px;color:#5146e5} .layout{display:grid;grid-template-columns:1fr 1.35fr;gap:20px} h2{font-size:19px;margin-top:0} label{display:block;font-size:13px;font-weight:bold;margin:14px 0 6px} input,select{width:100%;padding:12px;border:1px solid #ddd;border-radius:8px;font-size:14px;background:white} button{border:0;padding:11px 15px;border-radius:8px;background:#5146e5;color:white;font-weight:bold;cursor:pointer} button:hover{opacity:.88} .full{width:100%;margin-top:17px} .task{border:1px solid #eee;padding:14px;border-radius:10px;margin-top:11px;display:flex;align-items:center;gap:10px} .task input{width:18px;height:18px} .task-info{flex:1;min-width:0} .task-info strong{display:block;overflow-wrap:anywhere} .task-info small{display:block;color:#777;margin-top:6px;line-height:1.5} .done strong{text-decoration:line-through;color:#888} .delete{background:#ffe8e8;color:#c62828;padding:8px 10px} .progress-bg{height:12px;background:#e9e9f5;border-radius:10px;overflow:hidden;margin:12px 0} .progress-fill{height:100%;width:0;background:#5146e5;transition:width .3s} .badge{font-size:11px;background:#eeeaff;color:#5146e5;padding:5px 7px;border-radius:6px} .section{margin-top:20px} .empty{color:#888;text-align:center;padding:30px 10px} #message{font-size:13px;color:#5146e5;margin-top:10px} footer{text-align:center;padding:25px;color:#777;font-size:12px} @media(max-width:750px){.layout{grid-template-columns:1fr}header{padding:20px}} </style> </head>
```

Part 2: Semantic HTML Structure & Input Form Elements

<body> to <main>

```
<body> <header> <div> <h1>📚 Smart Study Planner</h1> <p>Plan smarter. Study better. Achieve more.
</p> </div> <div id="today"></div> </header> <main> <div class="stats"> <div class="card"><div class="sta
t-label">Total Study Tasks</div><div class="stat-number" id="total">0</div></div> <div class="card"><div
class="stat-label">Completed Tasks</div><div class="stat-number" id="completed">0</div></div> <div class
="card"><div class="stat-label">Pending Tasks</div><div class="stat-number" id="pending">0</div></div> <d
iv class="card"><div class="stat-label">Completion Rate</div><div class="stat-number" id="rate">0%</div>
</div> </div> <div class="layout"> <section class="panel"> <h2>✨ Create Study Schedule</h2> <p style="co
lor:#777;font-size:13px">Enter your subjects and available study time to create a smart plan.</p> <form i
d="studyForm"> <label for="subject">Subject Name</label> <input id="subject" placeholder="e.g. Python Pro
gramming" required> <label for="topic">Topic / Task</label> <input id="topic" placeholder="e.g. Functions
and Loops" required> <label for="date">Study Date</label> <input type="date" id="date" required> <label f
or="time">Preferred Start Time</label> <input type="time" id="time" value="17:00" required> <label for="d
uration">Duration</label> <select id="duration"> <option value="30">30 minutes</option> <option value="4
5">45 minutes</option> <option value="60" selected>1 hour</option> <option value="90">1.5 hours</option>
<option value="120">2 hours</option> </select> <label for="priority">Priority</label> <select id="priorit
y"> <option value="High">High</option> <option value="Medium" selected>Medium</option> <option value="Lo
w">Low</option> </select> <button class="full" type="submit">+ Add to Study Plan</button> <div id="messag
e"></div> </form> <button class="full" type="button" onclick="generatePlan()">⚡ Generate Smart Daily Pla
n</button> <button class="full" type="button" onclick="clearCompleted()" style="background:#777">Clear Co
mpleted Tasks</button> </section> <section class="panel"> <h2>📊 Study Progress</h2> <div style="display:
flex;justify-content:space-between;gap:10px"> <span>Overall completion</span> <strong id="progressText">
0%</strong> </div> <div class="progress-bg"><div class="progress-fill" id="progressBar"></div></div> <p s
tyle="color:#777;font-size:13px">Check off each task when you finish studying.</p> <div class="section">
<h2>🗓️ Your Study Schedule</h2> <label for="filter">Filter Tasks</label> <select id="filter" onchange="re
nderTasks()"> <option value="All">All Tasks</option> <option value="Pending">Pending Tasks</option> <opti
on value="Completed">Completed Tasks</option> </select> <div id="taskList"></div> </div> </section> </div
> </main> <footer>Smart Study Planner | Student Productivity Project</footer>
```

Part 3: Complete JavaScript Engine & Scheduler Logic

<script> logic

```
<script> const $ = id => document.getElementById(id); const today = new Date(); const localDate = d => { const y = d.getFullYear(); const m = String(d.getMonth()+1).padStart(2,"0"); const day = String(d.getDate()).padStart(2,"0"); return `${y}-${m}-${day}`; }; $("today").textContent = today.toLocaleDateString(undefined,{ weekday:"short",day:"numeric",month:"short",year:"numeric" }); $("date").value = localDate(today); $("date").min = localDate(today); let tasks = []; try { tasks = JSON.parse(localStorage.getItem("studyPlannerTasks") || "[]"); if (!Array.isArray(tasks)) tasks = []; } catch { tasks = []; } function saveTasks() { try { localStorage.setItem("studyPlannerTasks",JSON.stringify(tasks)); } catch { alert("Unable to save data in this browser."); } } function escapeHTML(value) { return String(value).replace(/[&<>"']/g, char => ({ "&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;","'":"&#39;" })[char]); } function formatTime(value) { if (!value) return ""; const [h,m] = value.split(":").map(Number); const d = new Date(); d.setHours(h,m,0,0); return d.toLocaleTimeString([],{ hour:"numeric",minute:"2-digit" }); } function endTime(time,duration) { const [h,m] = time.split(":").map(Number); const total = h*60+m+Number(duration); return String(Math.floor(total/60)%24).padStart(2,"0")+":"+String(total%60).padStart(2,"0"); } $("studyForm").addEventListener("submit",function(event){ event.preventDefault(); const subject = $("subject").value.trim(); const topic = $("topic").value.trim(); const date = $("date").value; const time = $("time").value; const duration = Number($("duration").value); const priority = $("priority").value; if (!subject || !topic || !date || !time) { $("message").textContent = "Please fill in all fields."; return; } tasks.push({ id: Date.now().toString()+Math.random().toString(36).slice(2), subject,topic,date,time,duration,priority,completed:false }); saveTasks(); renderTasks(); this.reset(); $("date").value = localDate(today); $("time").value = "17:00"; $("duration").value = "60"; $("priority").value = "Medium"; $("message").textContent = "Study task added successfully!"; }); function renderTasks() { const list = $("taskList"); const filter = $("filter").value; const visible = tasks.filter(t => filter === "All" || (filter === "Completed" && t.completed) || (filter === "Pending" && !t.completed) ).sort((a,b) => a.date.localeCompare(b.date) || a.time.localeCompare(b.time)); if (!visible.length) { list.innerHTML = '<div class="empty">No tasks here yet. Add a study task to get started!</div>'; } else { list.innerHTML = visible.map(t => ` <div class="task ${t.completed ? "done" : ""}"> <input type="checkbox" aria-label="Mark task completed" ${t.completed ? "checked" : ""} onchange="toggleTask('${t.id}')"> <div class="task-info"> <strong>${escapeHTML(t.topic)}</strong> <small>${escapeHTML(t.subject)}<br>${escapeHTML(t.date)} · ${escapeHTML(formatTime(t.time))} – ${escapeHTML(formatTime(endTime(t.time,t.duration)))}<br>${t.duration} min · Priority: ${escapeHTML(t.priority)} </small> <span class="badge">${t.completed ? "Completed" : "Pending"}</span> </div> <button class="delete" aria-label="Delete task" onclick="deleteTask('${t.id}')">X</button> </div>`).join(""); } const total = tasks.length; const completed = tasks.filter(t => t.completed).length; const rate = total ? Math.round(completed/total*100) : 0; $("total").textContent = total; $("completed").textContent = completed; $("pending").textContent = total-completed; $("rate").textContent = rate+"%"; $("progressText").textContent = rate+"%"; $("progressBar").style.width = rate+"%"; } function toggleTask(id) { const task = tasks.find(t => t.id === id); if (!task) return; task.completed = !task.completed; saveTasks(); renderTasks(); } function deleteTask(id) { if (!confirm("Delete this study task?")) return; tasks = tasks.filter(t => t.id !== id); saveTasks(); renderTasks(); } function clearCompleted() { if (!tasks.some(t => t.completed)) { alert("There are no completed tasks to clear."); return; } if (!confirm("Remove all completed tasks?")) return; tasks = tasks.filter(t => !t.completed); saveTasks(); renderTasks(); } function generatePlan() { const date = $("date").value || localDate(today); const dayTasks = tasks.filter(t => t.date === date); if (!dayTasks.length) { alert("First add some tasks for " + date + "."); return; } const priorityOrder = {High:0,Medium:1,Low:2}; dayTasks.sort((a,b) => priorityOrder[a.priority]-priorityOrder[b.priority] || a.time.localeCompare(b.time)); let current = Math.min(...dayTasks.map(t => { const [h,m] = t.time.split(":").map(Number); return h*60+m; })); for (const task of dayTasks) { task.time = String(Math.floor(current/60)%24).padStart(2,"0")+":"+String(current%60).padStart(2,"0"); current += Number(task.duration)+10; } saveTasks(); renderTasks(); $("filter").value = "All"; alert("Smart study plan generated! Tasks are arranged by priority with 10-minute breaks."); } renderTasks(); </script> </body> </html>
```